Field-Cooperative Neural Topic Modeling of Civil Engineering Literature: Read-out Evaluation and Missing-Abstract Inference

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ABSTRACT

Title–abstract records are a core form of civil engineering literature, and part of most collections is available with titles only. This paper studies how the two fields of such a record should be modelled together, and what can still be inferred when the abstract is missing. We present FC-NTM, a field-cooperative neural topic model with one conditional variational stream per field, cross-field reconstruction, a consistency term and field-level masking, evaluated on civil engineering records from OpenAlex together with two public short-text corpora. Under a unified read-out protocol the civil corpus shows the largest and most consistent advantage: across ten random seeds the class TF-IDF coherence is 0.049 and the native beta read-out 0.061 above BTM, with ten positive pairs. A title-only routing task reaches 0.66 accuracy on fifteen external automatic topic labels covering 42.9% of the test records; the task is a routing aid and remains below a supervised TF-IDF classifier. Cross-domain evaluations and read-out protocols are reported in the paper and supplement, and per-seed results, fixed data splits and hyper-parameters are released with the code.

Keywords: topic model; field cooperation; neural variational inference; fielded documents with short summary fields; class TF–IDF read-out; complementarity measures

INTRODUCTION

Topic models summarise unlabelled collections as readable topic structures and are widely used in news, software repositories and social media analysis. In many of these collections a

record is not a homogeneous bag of words but a pair of fields with different functions: a title that gives a highly compressed summary, and a body that supplies context, motivation and qualifications. The same topic is expressed differently in the two fields – the body side tends to carry context and causes, the summary side tends to carry objects and actions. Civil and construction engineering records are a natural instance: bibliographic title–abstract pairs, technical specifications and contract clauses, and accident reports all pair a short summary-like field with a longer contextual field. Topic modelling of such documents is useful both for literature navigation and for records whose long field is unavailable, for example when only titles have been collected.

Topic modelling has moved from probabilistic graphical models to neural variational models. LDA is the classical bag-of-words Bayesian model (Blei et al. 2003); GSDMM addresses short texts with a collapsed Gibbs Dirichlet–multinomial mixture (Yin and Wang 2014); AVITM introduces a linear decoder into a variational auto-encoder and defines the LDA and ProdLDA generative processes (Srivastava and Sutton 2017); NVDM applies variational auto-encoders to text early on (Miao et al. 2016); CTM/CombinedTM combine contextual sentence embeddings with bag-of-words counts (Bianchi et al. 2021); ETM defines topics in a word-embedding space (Dieng et al. 2020); BTM models word pairs instead of document-level counts and remains a strong baseline for short texts (Yan et al. 2013). Surveys note that structured and short-text corpora are still open problems for neural topic models (Wu et al. 2024).

Existing neural topic models exploit the internal field structure of a document only to a limited degree: the fields are usually concatenated into one bag of words and their division is discarded. Multi-view and multi-source topic modelling has been explored for parallel corpora, multiple data sources and cross-modal settings (Kurle et al. 2019; Zhang et al. 2017; Pergola et al. 2021; Liu et al. 2022); structured neural topic models for reviews encode user–item context and assign words to sentence-level aspects (Esmaeili et al. 2019); structural topic models describe metadata with document-level covariates (Roberts et al. 2019). Short-text

work such as DV-BTM optimises two views jointly (Liu et al. 2022), CRNTM reinforces topic inference with context (Feng et al. 2022), and DVI-DTM learns dual-view representations for interpretable dynamic topics (Liu et al. 2026). In these works the two views play known roles (query–document, user–item, or aspect layers); the setting studied here differs: the two fields of the *same* record are modelled by separate variational streams, and their relation is fixed by the record schema rather than inferred as an explicit latent variable. Separate encoders, posterior alignment and missing-view inference are building blocks adopted from this literature; the contribution claimed here is the field-cooperative configuration and its read-out protocol, not those building blocks themselves.

In engineering informatics, text mining has become a recognised analysis tool: content analysis and classification of safety reports (Tixier et al. 2016; Zhang et al. 2019), automated classification of regulations and contracts and extraction of contractual risk clauses (Salama and El-Gohary 2016; Zhou and El-Gohary 2016; Moon et al. 2022; Pham and Han 2023), and surveys of natural language processing for construction (Wu et al. 2022; Shamshiri et al. 2024). Topic models have entered the field as well: DUET applies LDA topic modelling to social-media text for detecting physical events in urban areas (Wang and Taylor 2019), neural topic modelling has been used to map machine-learning applications in buildings (Zhou et al. 2025), and semantic topic modelling has been used to build taxonomies for digital building passports (Vangelova et al. 2026). In contrast to these works, we do not treat the topic model as a feature extractor for a downstream classifier. We study the *internal* field structure of an engineering record: topic inference when a field is missing, field-restricted read-out, and whether these read-outs can be measured reliably.

This paper makes two contributions. **Methodologically**, it proposes the field-cooperative neural topic model FC-NTM, which models the two fields of a record as two conditional variational streams that share one linear topic decoder, adds cross-field reconstruction and a field consistency regulariser, and handles partial and complete field absence with field-level word masking and whole-field dropout; a single trained model yields both a decoder β read-out

and a class TF–IDF read-out under a unified evaluation vocabulary. **Empirically**, the paper does not claim that the mechanism beats classical baselines on arbitrary dual-field corpora. It asks what field structure provides and whether the resulting gains can be measured reliably. On the civil corpus and on GitHub issues (background/summary length ratio 12.9) and AG News (3.9) we extend FC-NTM and BTM to ten seeds: both read-outs favour FC-NTM on the civil corpus (+0.049 class TF–IDF and +0.061 β C_V, 10/10 seeds), the class TF–IDF read-out is +0.028 C_V above BTM on GitHub (95% CI $[+0.012, +0.043]$, 9/10 seeds) and -0.009 on AG News (CI includes zero, 3/10), while on the two public corpora the β read-out is lower than BTM. We therefore claim the multi-seed-supported read-out advantages on the civil corpus and on GitHub, and report the absence of an advantage on AG News. Two field-relation measures are tested as candidate predictors – a held-out mutual-prediction score and a post-training occlusion diagnostic – and neither serves as a monotone criterion. The value of field cooperation is thus restricted to reproducible field-restricted read-outs and to the behaviour of inference under missing fields, not to an unconditional performance advantage.

RELATED WORK

Neural topic models

Neural topic models are mostly built on variational auto-encoders: an encoder maps the document bag of words to the parameters of a latent Gaussian posterior and a decoder generates word distributions. NVDM applied variational auto-encoders to text (Miao et al. 2016), ProdLDA introduced a linear decoder with a readable topic–word matrix (Srivastava and Sutton 2017), CTM/CombinedTM concatenate pretrained sentence embeddings with bag-of-words counts (Bianchi et al. 2021), and ETM defines topics in a continuous embedding space (Dieng et al. 2020). For short texts, BTM models word pairs rather than document-level counts (Yan et al. 2013), while BERTopic and Top2Vec build topic representations from class TF–IDF statistics and joint embeddings respectively (Grootendorst 2022; Angelov 2020). Neural and regularised variants of BTM include GraphBTM (Zhu et al. 2018) and BCK-TM, which encodes word-pair correlation knowledge into the prior (Zhang

et al. 2022); BigARTM provides a general additive regularisation framework for multimodal topic modelling (Vorontsov et al. 2015). Our decoder follows the linear ProdLDA structure and adds topic orthogonality, dropout (Srivastava et al. 2014) and validation-based early stopping (Prechelt 1998); short-text-specific decoders use topic quantisation and negative sampling to sharpen assignments (Wu et al. 2020). We keep this regularisation consistent across the neural baselines we implement ourselves, while the official CombinedTM keeps its own training loop; learning rates and KL weights differ between models, so the training objectives should not be described as identical.

Multi-view and structured-document topic models

Multi-view topic models share information between views of the same data: bilingual and cross-lingual models, multi-source models and disentangled representation models (Kurle et al. 2019; Zhang et al. 2017; Pergola et al. 2021; Liu et al. 2022). Structured neural topic models for reviews encode user–item context and sentence-level aspect assignments (Esmaeili et al. 2019), neural models can also condition on metadata (Card et al. 2018), and the structural topic model conditions topics on document metadata (Roberts et al. 2019). Short-text-oriented dual-view work is closest to ours: DV-BTM learns two views jointly (Liu et al. 2022), CRNTM strengthens inference with context (Feng et al. 2022), and DVI-DTM combines a time-aware term view with a sentence view and adds interpretable topic descriptions (Liu et al. 2026); related word-pair models address relations and sparsity (Li et al. 2019; Zhu et al. 2021). Two further lines overlap partially. Multi-source variational inference with per-source encoders, a shared latent code and divergence terms between posteriors (Kurle et al. 2019) and JMVAE, which aligns modality posteriors and can infer a missing modality from the remaining one (Suzuki et al. 2016), already contain separate encoders, posterior alignment and missing-view inference; SubNTM instead models a hierarchy of document and sub-document topics (Nguyen et al. 2026), i.e. structure *inside* a field. The setting studied here is defined by the record schema rather than by unknown view roles: two textual fields of the same record share one bag-of-words vocabulary and one linear decoder, the fields enter

with fixed weights, and the evaluation asks what the field-restricted read-outs and missing-background inference measure. Separate encoding, posterior alignment and missing-view handling are therefore building blocks adopted from this literature rather than contributions claimed as new here.

Text mining and topic modelling of engineering documents

Engineering informatics has accumulated a substantial body of text-mining work: content analysis and classification of construction safety reports (Tixier et al. 2016; Zhang et al. 2019), automated classification of regulatory and contractual documents and extraction of risk clauses (Salama and El-Gohary 2016; Zhou and El-Gohary 2016; Moon et al. 2022; Pham and Han 2023), and domain surveys of natural language processing for construction (Wu et al. 2022; Shamshiri et al. 2024). Topic models have also been applied: DUET uses LDA topic modelling on social-media text to detect physical events in urban areas (Wang and Taylor 2019), recent work applies neural topic modelling to map machine-learning applications in buildings (Zhou et al. 2025), and semantic topic modelling supports taxonomy construction for digital building passports (Vangelova et al. 2026). These studies use topics as inputs to downstream classification or as a mapping tool. We instead study the field structure inside a single engineering record – title versus abstract – and ask when field-restricted topics and missing-field inference are informative, and when the corresponding gains are measurable.

METHOD

Problem setting

Each record d consists of field A (*background*, e.g. the abstract of a paper or the body of an issue) and field B (*summary*, e.g. the title), with N records in total and a shared evaluation vocabulary of V terms. The model maintains K topics, a document–topic distribution $\theta_d \in \Delta^{K-1}$, and a topic–word matrix $\beta \in \mathbb{R}^{K\times V}$ shared by both fields. The task is to infer θ_d and β from unlabelled records, to produce readable topics in two read-outs, and to remain usable when one field is partially or entirely missing at inference time.

Field streams and the shared decoder

Each field is encoded by its own two-layer MLP encoder (200 hidden units, Softplus activations, dropout 0.2) into the parameters of a field-conditioned Gaussian posterior,

$$q_f(z_f \mid x_f) = \mathcal{N}\big(\mu_f(x_f), \mathrm{diag}(\sigma_f^2(x_f))\big), \qquad f \in \{\mathrm{bg}, \mathrm{sm}\}, \tag{1}$$

where x_f is the L_1-normalised bag of words of field f and $\log \sigma_f^2$ is capped at 2. Posterior means are mapped to topic proportions by a deterministic softmax, $\bar{\theta}_f = \mathrm{softmax}(\mu_f(x_f))$, which is used at evaluation time in place of sampling; $\bar{\theta}_f$ is therefore a deterministic topic-proportion estimate, not the expectation of $\mathrm{softmax}(z_f)$. Both streams share one linear topic–word matrix $W \in \mathbb{R}^{K \times V}$, and write $\theta_f \in \mathbb{R}^{1 \times K}$ for the topic proportions of field f under the row-vector convention,

$$p(w \mid \theta_f) = \mathrm{softmax}\big(\mathrm{BN}(\theta_f W)\big), \tag{2}$$

where θ_f passes through dropout before the product, BN is a batch-normalisation layer *without* affine parameters applied over the vocabulary dimension (batch statistics during training, running statistics at evaluation), and there is no additive bias. The β read-out of the "Unified topic-word read-out" subsection reads the raw rows of W, whereas generation uses the normalised path of Eq. (2); the gap between the two is quantified in the Supplementary Material.

Training objective

For a document d let $\mathcal{G}_d$ collect its present fields (the background field is present when the record carries it; the summary stream is always instantiated, including zero-vector inputs after preprocessing) and let $\mathcal{A}_d \subseteq \mathcal{G}_d$ be the *active streams*, i.e. present fields whose stream is not removed by whole-field dropout. Write $n_{dw}^{(g)}$ for the raw count of word w in field g.

Each active stream decodes once (Eq. (2)) and reconstructs *every present field*,

$$R_f = \sum_{g \in \mathcal{G}_d} \omega_g \, R_f(g), \qquad R_f(g) = -\sum_{w=1}^{V} n_{dw}^{(g)} \, \log p(w \mid \theta_f) \,, \tag{3}$$

with fixed field weights $\omega_{\text{bg}} = 2.0$ and $\omega_{\text{sm}} = 1.0$. The summary stream therefore reconstructs the background field as well (cross-field reconstruction), and a background field that was observed but dropped during training stays in the reconstruction targets even though its own stream is inactive, whereas a naturally missing background field enters neither the inputs nor the targets. The per-document objective is

$$\mathcal{L}_d = \frac{1}{|\mathcal{A}_d|} \sum_{f \in \mathcal{A}_d} \Big[R_f + \lambda_{\text{KL}}(t) \, \text{KL}_f \Big] + \lambda_f R_{\text{fuse}} + \lambda_{\text{JS}} \, \text{JS}, \tag{4}$$

where $\text{KL}_f = \text{KL}\big(q_f(z_f \mid x_f) \,\|\, \mathcal{N}(0, I)\big)$ is summed over the K latent dimensions and clamped at zero, and $\lambda_{\text{KL}}(t) = 0.05 \min(1, t/20)$ anneals linearly during the first 20 epochs. The fused term $R_{\text{fuse}} = \sum_{g \in \mathcal{G}_d} \omega_g R_{\text{fuse}}(g)$ is a second reconstruction of both present fields from the fused topic mixture

$$\theta_d^{\text{fuse}} = g \, \theta_{\text{bg}} + (1 - g) \, \theta_{\text{sm}}, \qquad g = \sigma(w), \quad w \in \mathbb{R}, \tag{5}$$

for records with both fields present, and $\theta_d^{\text{fuse}} = \theta_{\text{sm}}$ when the background field is missing or dropped. During training Eq. (5) combines the *sampled* per-stream proportions that also feed the per-stream reconstructions; at inference time, and for the JS term below, the deterministic estimates $\bar{\theta}_f$ are used instead, so at inference Eq. (5) is a convex combination of two deterministic topic-proportion estimates, which is why we call it a *fused topic mixture* rather than a posterior. The gate g receives gradients only through R_{fuse}, so with $\lambda_f = 0$ the model degenerates exactly to fixed mean fusion. The consistency term

$$\text{JS} = \tfrac{1}{2} \, \text{KL}\big(\bar{\theta}_{\text{bg}} \| m\big) + \tfrac{1}{2} \, \text{KL}\big(\bar{\theta}_{\text{sm}} \| m\big), \qquad m = \tfrac{1}{2}\big(\bar{\theta}_{\text{bg}} + \bar{\theta}_{\text{sm}}\big), \tag{6}$$

acts on the deterministic topic proportions, not on the Gaussian posteriors, and is computed only for documents whose two streams are both active, so no record is implicitly regularised more strongly than another. A topic-orthogonality penalty keeps the rows of W apart,

$$\Omega = \frac{1}{K(K-1)} \sum_{k \neq k'} \left| \cos\langle W_k, W_{k'} \rangle \right|, \tag{7}$$

and the batch objective is $\frac{1}{|B|} \sum_{d \in B} \mathcal{L}_d + \lambda_{\text{orth}} \Omega$ with $\lambda_{\text{orth}} = 0.2$.

Two masking mechanisms expose the model to missing fields, and they act on different units. (i) *Word-level input dropout* (denoising) is applied during training to the bag-of-words *word types* of each field: each BoW dimension is independently zeroed with probability 0.30 (background) or 0.40 (summary) in the encoder view only, while the reconstruction targets remain the complete counts. (ii) *Whole-field dropout* removes the background stream of each training document with probability 0.5 and substitutes the prior; the summary stream is never dropped and learns to reconstruct both fields on its own. The test-time occlusion protocol of the "Field inference under partial occlusion" subsection instead masks *token positions* in the raw field text, so the training and evaluation units should not be conflated. The consistency weight $\lambda_{\text{JS}} = 0.5$, the fusion weight $\lambda_f = 0.5$ and the dropout probabilities were selected on held-out data.

Optimisation

Training uses Adam (learning rate 5×10^{-4}, momentum parameters 0.9 and 0.999), mini-batches of 128, gradient-norm clipping at 5.0 and a budget of 700 epochs. After the KL warm-up the state with the lowest held-out loss is kept, for which 20% of the training split is held out; the patience is 150 epochs. Evaluation-mode inference is deterministic: the reparameterisation sampling is switched off and the held-out loss uses posterior means. Numerical-anomaly recovery rolls the weights back to the most recent finite epoch, clears the optimiser moments and halves the learning rate; this recovery step is separate from the best-held-out-epoch selection of the early-stopping loop.

Unified topic-word read-out

Topic-word comparisons are only meaningful if every model is read out with the same vocabulary and the same number of words per topic. We therefore fix one evaluation vocabulary and exactly ten words per topic. The β **read-out** takes the top ten words of each normalised row of the model's topic–word matrix; for FC-NTM these are the raw rows of W, whereas generation uses Eq. (2). LDA, BTM, NMF, ProdLDA, ETM, CTM-lite, CombinedTM and FC-NTM are all evaluated this way. The **class TF–IDF read-out** aggregates the training-term counts by the soft document–topic assignment of the model, applies a BERTopic-style class TF–IDF transform, and then takes the top ten words; for FC-NTM this read-out is built from the fused topic mixture $\bar{\theta}^{\text{fuse}}$, which is why the two read-outs can rank models differently. The sensitivity of the raw-parameter read-out relative to the generative read-out is reported in the Supplementary Material.

EXPERIMENTS

Corpora and evaluation protocol

We use three dual-field corpora. The **civil engineering corpus** is the application corpus: title–abstract records of English journal articles from OpenAlex (Priem et al. 2022) in *Civil and Structural Engineering* and *Building and Construction* (2000–2023, CC-licensed records only), 6000 training and 2000 test records, with the title as the summary field and the abstract as the background field (mean length ratio 11.2). **GitHub issues** (open index 2026) pairs the issue body with its title (four mature repositories, 6000/2000, ratio 12.9), and **AG News** (Zhang et al. 2015) pairs the description with the headline (6000/7600, ratio 3.9). All corpora are lower-cased and tokenised on letter sequences; stop words and very short tokens are removed, the vocabulary is built from the training split only, and test records never enter the vocabulary. Table 1 reports the application corpus and Tables 2 and 3 the two public corpora, all on the same evaluation vocabulary with exactly ten words per topic and $K = 30$. The evaluation protocol (vocabulary, ten words per topic, coherence and diversity metrics) is shared by every model, whereas the training procedure differs across algorithm families

and is described in the "Baselines, training protocols and metrics" subsection. Because a 400-record development subset was used during design, every result is also recomputed on a development-free test subset (1600 GitHub and 7200 AG News records); the re-scoring removes direct test overlap only and cannot undo the model selection that already happened, so it is reported as a sensitivity analysis rather than as an untouched confirmation set. The results are organised around three questions: how well civil title–abstract records can be structured into verifiable topics; how much signal remains when the abstract is missing; and how much the complete field-cooperative configuration adds over a concatenated single-encoder control. Cross-corpus diagnostics support these questions rather than replacing them.

Baselines, training protocols and metrics

Baselines are LDA (Blei et al. 2003), BTM (Yan et al. 2013), NMF with a KL objective (Lee and Seung 2001), ProdLDA (Srivastava and Sutton 2017), ETM (Dieng et al. 2020), CTM-lite, the official CombinedTM (Bianchi et al. 2021), BERTopic (Grootendorst 2022) and Top2Vec (Angelov 2020); contextual baselines use Sentence-BERT embeddings (Reimers and Gurevych 2019). The implementations fall into four families whose training procedures differ, and they are not interchangeable training loops. (i) *Gibbs sampling*: BTM is a collapsed Gibbs sampler over biterms (word pairs within a window of 12 tokens, at most 150 sampled pairs per document, $\alpha = 50/K$, $\beta = 0.01$, 300 full sweeps); LDA is fitted by gensim's batch variational Bayes (50 passes, 120 iterations per pass, symmetric α). (ii) *Matrix factorisation*: NMF minimises the Kullback–Leibler divergence between the count matrix and WH with multiplicative updates (600 iterations, tolerance 10^{-4}). (iii) *Neural variational models*: ProdLDA, ETM, CTM-lite and FC-NTM share the same Adam loop, mini-batch size and budget (700 epochs, 20% held-out early stopping), but the learning rate differs – the baselines use 2×10^{-3}, the main FC-NTM configuration uses 5×10^{-4} selected on development data – and the official CombinedTM keeps its own training loop (Adam, learning rate 2×10^{-3}, batch size 64, 700 epochs, early stopping on its own held-

out split). (iv) *Embedding-clustering models* choose their own topic numbers: BERTopic runs UMAP followed by HDBSCAN with fixed random seeds, and Top2Vec fits Doc2Vec document vectors followed by HDBSCAN; both are reported for reference only. Quality is measured by C_V coherence (Röder et al. 2015), NPMI (Bouma 2009; Lau et al. 2014), UCI and UMass, IRBO topic diversity (Webber et al. 2010) and a K-corrected exclusivity $EXCL_{KADJ}$ (Bischof and Airoldi 2012); coherence scores drop topics with non-finite scores, so we record the effective topic count next to every score. Definitions of the two corrected measures are given in the Supplementary Material.

Main results

On the civil engineering corpus (Table 1), FC-NTM attains the highest coherence among all evaluated models under both read-outs ($C_V = 0.633$ for the β read-out and 0.636 for the class TF–IDF read-out; NPMI 0.154 and 0.159 over seeds 11/23/37), ahead of NMF (0.598), BTM (0.583) and the neural baselines, and the ten-seed extension keeps both read-out advantages (Table 5). On AG News, FC-NTM-cTFIDF reaches $C_V = 0.612$ and NPMI $= 0.173$, above BTM under the matched class TF–IDF read-out (0.593 / 0.169) and clearly above the β read-outs of ProdLDA, ETM, NMF, CTM-lite and LDA; under its own β read-out BTM keeps a small lead (0.600 versus 0.586). On GitHub issues, FC-NTM-cTFIDF reaches $C_V = 0.727$ and NPMI $= 0.274$, above BTM at both read-outs (0.712 / 0.232 for the native β read-out, 0.700 / 0.235 matched), and above the sentence-embedding baselines (CombinedTM 0.643, BERTopic 0.540). The matched-read-out advantage over BTM is therefore +0.027 C_V and +0.039 NPMI on GitHub and +0.019 / +0.004 on AG News for three seeds, but the per-seed structure differs: on GitHub all three seeds are positive, whereas the AG News mean is carried by two of three seeds and the NPMI advantage by one. On the development-free test subset the ordering is unchanged (GitHub 0.734 versus 0.711, NPMI 0.284 versus 0.248; AG News 0.610 versus 0.590, NPMI 0.172 versus 0.167; with BTM's native β read-out, 0.734 versus 0.726 and 0.610 versus 0.598).

Read-out-matched comparison

The class TF–IDF read-out is strong but its effect depends on the model and the corpus: on GitHub it lifts CTM-lite, ProdLDA and LDA by 0.091, 0.087 and 0.068 but leaves ETM (-0.006) and NMF (-0.011) unchanged or worse, while on AG News it helps every model except BTM by $+0.005$–$+0.045$. Comparing FC-NTM-cTFIDF against baselines read out with their own β matrix therefore mixes the model and the read-out; every model-quality statement below therefore uses the matched comparison. Under the matched read-out the strongest baselines on GitHub are BTM (0.700), CTM-lite (0.675), ETM (0.669) and ProdLDA (0.666), and FC-NTM-cTFIDF leads them by 0.027–0.061 C_V and 0.039–0.059 NPMI, larger than its seed-to-seed standard deviation (0.022). On AG News the matched advantage is $+0.019$ C_V over BTM but only two of three seeds are positive and the NPMI advantage rests on one seed, while under the native β read-out FC-NTM is below BTM (0.586 versus 0.600). On the civil corpus the strongest matched-read-out baselines are NMF (0.607) and ETM (0.603), ahead of BTM (0.590); FC-NTM-cTFIDF leads them by 0.029 and 0.033 C_V, but the ten-seed paired comparison is against BTM, so the multi-seed civil evidence is specific to that comparison. The stable gains are therefore concentrated on the matched read-out, with a corpus-specific pattern. Two caveats apply: the class TF–IDF transform does not automatically favour BTM (its C_V drops from 0.712 to 0.700 on GitHub and from 0.600 to 0.593 on AG News), so its direction must be measured rather than assumed; and the architecture hyper-parameters of FC-NTM were selected on development data while the baselines use their defaults, a tuning-budget difference that the read-out comparison cannot remove.

Ten-seed paired comparison

To test whether the three-seed differences are seed noise, we extend FC-NTM and BTM to ten seeds on all three corpora (adding 53/71/89/101/113/127/139) and compare both read-outs and the occlusion behaviour seed by seed; the occlusion rows use pre-selected baselines (BTM on all three corpora and additionally LDA on AG News) rather than a full

tournament.

Four observations follow. First, the matched read-out advantage on GitHub survives ten seeds: FC-NTM-cTFIDF is +0.028 C_V above BTM (95% CI $[+0.012, +0.043]$, 9/10 seeds positive) and its relative degradation advantage is +0.024 (10/10, $p = 0.002$). Second, the civil corpus gives the largest and most consistent read-out advantage: +0.049 class TF–IDF and +0.061 native β C_V (the latter refers to the raw-parameter read-out; the BN-generated counterpart is smaller, see the Supplementary Material), both with all ten paired differences positive ($p = 0.002$), although the occlusion ratio still favours BTM (-0.003, 0/10). Third, the three-seed AG News advantage is not replicated under ten seeds: the matched read-out difference is -0.009, with a confidence interval that includes zero (3/10). Fourth, on the two public corpora the native β read-out is below BTM (-0.014 and -0.027); overall the advantage is corpus-dependent and read-out-dependent: largest on the civil corpus under both read-outs, moderate and class-TF–IDF-focused on GitHub, and neutral on AG News. Confidence intervals describe training (and masking) randomness under a fixed corpus and a fixed design; the p-values are exploratory and not corrected for multiple comparisons.

Field redundancy and the gain

The redundancy measures are fitted on the training split and evaluated on the test split. The two corpora have almost the same held-out predictability (GitHub is slightly higher, ρ^2_{cca} 0.306 versus 0.245), while FC-NTM-cTFIDF is slightly above BTM on both at three seeds (+0.027 and +0.019 matched) and the ten-seed extension keeps the difference on GitHub (+0.028, 9/10) and turns it negative on AG News (-0.009, 3/10); the civil corpus, whose redundancy is not measured here, shows the largest extension advantage (Table 5). The evidence therefore supports neither "complementary fields gain" nor "redundant fields pay a price". The differences (0.019–0.027 matched, 0.012–0.014 against BTM's native β read-out) are of the same order as the seed-to-seed variation, so they are described as a small advantage over the strongest classical baseline rather than as a significant improvement. The occlusion diagnostic points in the same direction: both corpora change coherence by almost

nothing (GitHub $+0.009 \pm 0.022$, AG News $+0.018 \pm 0.013$). We therefore do not use field redundancy as a predictor of the gain; establishing such a criterion would require calibration on more corpora and an independent test corpus.

Field inference under partial occlusion

The central mechanism is field complementarity: when part of field A is unavailable, a model should use the still visible field B to recover the missing content. The protocol (Supplementary Material) masks a fraction p of the background-field tokens in the test documents, re-infers the topic proportions from the remaining input, and scores the occluded tokens under the model's own word distribution; it reports the per-token NLL, the agreement of the most probable topic with the unoccluded inference, and the posterior JS divergence. Every model uses its own inference path and its own word distribution, so no cross-model read-out convention is involved. When a document has no visible evidence at all, BTM and NMF fall back to the training-corpus topic prior instead of treating a zero vector as a distribution, which would inflate the NLL to $\log 10^{12} = 27.63$.

From 25% to 100% occlusion the per-token NLL of FC-NTM rises by only 3.2% ($7.712 \rightarrow 7.960$), less than LDA (7.4%), ProdLDA (8.0%), ETM (8.9%), BTM (5.2%) and NMF (47.1%), which is consistent with the training-time whole-field dropout. Absolute prediction and topic stability, however, are not led by FC-NTM: at full occlusion BTM has the lowest NLL (6.972) and the highest agreement belongs to NMF (0.469) and ETM (0.452), while FC-NTM reaches 0.358. We therefore present the relative degradation as a stability indicator, complemented by the absolute NLL and the agreement metric for missing-field prediction. Three qualifications hold: the ratio summarises each model's relative increase in NLL from the 25% occlusion condition, and is read together with the absolute NLL and the shuffled-summary control, which carry the calibration and input-sensitivity information; the protocol covers three corpora with three seeds and masks redrawn per seed, but all corpora are English and the field split follows existing title/body structure; and stability can also arise from insensitivity to the input, because the protocol occludes the background field itself.

Cross-corpus comparison

Three conclusions follow. First, the occlusion advantage is measured on only one of the three corpora and does not follow the length ratio: on GitHub (ratio 12.9) all models degrade measurably ($\times 1.028$–$\times 1.470$) and FC-NTM degrades least, beating the best baseline BTM by $+0.024 \pm 0.004$ in the ten-seed paired comparison (10/10, $p = 0.002$); on the civil corpus, whose ratio is almost identical (11.2), FC-NTM (1.010) leaves BTM marginally ahead on the same ratio (1.007 versus 1.010; paired difference -0.003, 0/10, $p = 0.002$); on AG News (3.9) LDA remains the reference to beat (1.006 versus 1.019). Second, no training-free statistic agrees with the three corpora: held-out mutual prediction is slightly higher for GitHub, and the length ratio and background-exclusive share separate GitHub from AG News but not from the civil corpus. Third, we therefore report these statistics as corpus descriptions rather than as a pre-training criterion; establishing such a criterion requires calibration on more corpora and an independent test corpus.

Field-restricted read-out

Field-restricted read-out truncates the topic words to one field's vocabulary and scores them against that field's own test text. With the shared decoder, the background-view C_V of FC-NTM stays close to its full-vocabulary β read-out on all three corpora (0.697, 0.566 and 0.637 versus 0.701, 0.586 and 0.633), while the summary view is lower in absolute terms mainly because a summary contains only 3–5 tokens and the reference corpus is very short, so that column should be compared within the same protocol only. The cross-model comparison shows that the read-out is not unique to a two-stream encoder: BTM is slightly better in the background view on GitHub and AG News (0.710 versus 0.697 and 0.586 versus 0.566), FC-NTM leads only on the civil corpus (0.637 versus 0.600 for NMF), and in the summary view LDA and ProdLDA are above FC-NTM on all three corpora (0.486 and 0.438 versus 0.423 on AG News). We therefore treat the field view as a descriptive read-out available to any model with a topic–word matrix, not as evidence of a unique structural advantage.

Ablation: fusion reconstruction and gating

The three-step ablation separates two questions: whether the fused mixture receives its own reconstruction term, and whether fusion is averaged or gated. The fusion-reconstruction term is effective, but only for the read-out that uses the fused posterior: adding it under mean fusion raises the class TF–IDF C_V and NPMI by 0.014 and 0.013 (3/3 seeds) while lowering the β read-out by 0.013 and 0.021. Replacing mean fusion by a gate brings no further gain in this corpus and protocol (class TF–IDF differences -0.001 and -0.000, 1/3 seeds; β read-out -0.007 and -0.012, 0/3). This conclusion has a precise boundary: without the fusion reconstruction term the gate receives no gradient and the model degenerates exactly to mean fusion, so the experiment cannot separate the value of the gate parameters from the value of the fusion objective; we report the absence of an observed gain rather than claiming that a trained gate is useless in general.

Ablation: core components

Four components are switched off one at a time. First, removing the JS consistency term causes no consistent change in topic quality on any corpus or read-out (all six cells $|\Delta C_V| \leq 0.004$, none with a clear positive paired majority): in this parameterisation the fused representation is trained by the fusion-reconstruction objective, not by pulling the two posteriors together. Second, the observed effect of cross-field reconstruction and word masking depends on the corpus: on GitHub, removing cross-field reconstruction even raises the β read-out by 0.021 (2/3 seeds) and costs only 0.005 in the class TF–IDF read-out, while on AG News and the civil corpus both read-outs drop when either component is removed (cross-field reconstruction $-0.011/-0.022$ and $-0.022/-0.019$; word masking $-0.026/-0.043$ and $-0.010/-0.012$, with all class TF–IDF columns at 0/3). Third, whole-field dropout acts mostly on the class TF–IDF read-out and is consistently beneficial on two of the three corpora only: it lowers both read-outs on GitHub and AG News but changes the β read-out of the civil corpus by only $+0.001$ while lowering its class TF–IDF read-out by 0.034. Fourth, field asymmetry does not explain the component-level differences: the

civil corpus has almost the same length ratio as GitHub (11.2 versus 12.9) yet behaves like AG News (3.9). The scope of the ablation is also limited: it switches off the per-stream cross-field reconstruction while the fused reconstruction term still rebuilds both fields, so it does not cut all information flow between fields.

Sensitivity to the fusion weight

At $\lambda_f = 0.25$ the class TF–IDF read-out is essentially the no-fusion result (C_V 0.713 versus 0.714) while the β read-out has already dropped by 0.024; between 0.5 and 1.0 the class TF–IDF gain is stable (0.727 versus 0.723, NPMI 0.274 versus 0.277, differences below 0.005), and the β penalty narrows from -0.020 to -0.005 relative to $\lambda_f = 0$. The value 0.5 used throughout is therefore a conservative compromise.

Whole-field occlusion

Deleting the background field entirely leaves topic-word coherence almost unchanged (all six cells within ± 0.02, with sign flips across seeds on GitHub), so coherence is insensitive to field loss and the information loss must be read from the document-level posterior and the word-level prediction: about 66% (GitHub) and 48% (AG News) of the documents change their most probable topic, and the per-token NLL rises by 0.24/0.23 with dropout training (0.24/0.27 without). Training with whole-field dropout changes the shape of the posterior rather than the calibration of word prediction: it lowers the per-token increase from 0.269 to 0.232 and the posterior JS from 0.060 to 0.049 on AG News, is directionally the same on the civil corpus (0.125→0.103, JS 0.039→0.033), and shows no increase difference on GitHub (0.242 versus 0.241) where only the JS slightly improves. Two corpora therefore show reduced posterior drift under field loss, which cannot replace absolute prediction accuracy.

Concatenated-field single-encoder control

To compare the full method with a single-encoder alternative under the same budget we train a control on the concatenated bag of words $[x_{\mathrm{bg}}; x_{\mathrm{sm}}]$ with the same shared linear decoder, field weights, KL warm-up and orthogonality penalty; whole-field dropout is emulated

by zeroing the background half of the input, and there is no fusion or consistency term. The concatenated input doubles the first encoder layer, keeping the parameter count comparable (a 1.5% difference), and both models use the same budget, vocabulary and read-outs on all three corpora (seeds 11/23/37).

The control is competitive everywhere, so the two-stream structure is not a necessary condition for topic quality. After the validation-mode correction the mean difference favours FC-NTM in every corpus–read-out cell (+0.010 to +0.029 C_V), but the paired majorities are mixed (1/3 to 3/3) and the gaps remain at the scale of the seed-to-seed variation with three seeds per cell, so the two-stream architecture is reported as a matched design choice rather than a source of unconditional coherence gains. The control also supports field-restricted read-out and title-only inference, so those read-outs are not exclusive to the two-stream design either.

Title-only routing on the civil corpus

When the abstract of a record is missing, can its title still be routed to an interpretable civil subfield? The evaluation labels are the OpenAlex primary topics of each record — external automatic labels from a classifier rather than expert ground truth; the frozen snapshot (topic IDs, fetch date) is released with the code, and the label set is the 15 most frequent topics of the training split, covering 42.9% of the test records (2000-record test split). Every model infers a topic distribution from the title only (FC-NTM and the control use their missing-background inference path); the per-model topic-to-label mapping is a supervised step — the majority training label of each argmax topic, with the training-label prior as fallback for topics that take no training record (three test records in one run triggered this fallback) — fitted on the same title-only inference, so the pipeline is an unsupervised representation with a supervised training-split mapping. Accuracy and macro-F1 are reported per seed; a supervised TF–IDF logistic regression on the same titles serves as a reference point, and FC-NTM with the full fields as an upper reference.

Topic-based routing from titles alone recovers a substantial part of the full-field signal

(0.664 versus 0.719 accuracy) and clearly outperforms the classical topic-model baselines (LDA 0.532, BTM 0.600), but it remains well below a supervised TF–IDF classifier trained on the same titles (0.810); the topic representation is learned without labels, while the topic-to-label mapping uses training labels, which is why the supervised classifier retains its advantage. The concatenated control reaches 0.653 against 0.664 for FC-NTM, but the three paired per-seed differences are not consistent in direction and the gap is small relative to the per-seed variation, so the two architectures operate within comparable ranges on this task. The task is a routing aid under a fixed label set, not a substitute for supervised classification, and its accuracy depends on the 42.9% label coverage of the 15-topic set.

DISCUSSION AND LIMITATIONS

The value of dual-field topic modelling has to be stated in parts. What is reproducible is the behaviour at the *representation* level: one trained model yields field-restricted topic words, and it still infers topic proportions when the background field is missing (on GitHub about 66% of the documents change their most probable topic and the per-token NLL rises by 0.24). What is corpus-dependent is the advantage on *generic* coherence metrics: under the matched read-out, the ten-seed comparison supports FC-NTM-cTFIDF being $+0.028$ C_V above BTM on GitHub (95% CI excludes zero, 9/10 seeds) and $+0.049$ on the civil corpus (10/10) but -0.009 on AG News (3/10); on the civil corpus the native β read-out is also above BTM ($+0.061$, 10/10). This differs from the simple hypothesis that complementary corpora benefit and redundant ones suffer, for two reasons: coherence metrics are sensitive to implementation details such as the decoding path, the training budget and Monte-Carlo noise, so measuring under one protocol is more reliable than extrapolating from intuition; and the held-out redundancy measures show that GitHub is in fact slightly the more predictable of the two corpora. Redundancy is therefore reported as a description of the field relation, not as a predictor of the direction of the gain.

For engineering document management the protocol has direct uses: the field-restricted read-out matches the way titles and abstracts are actually written, the occlusion diagnostic

corresponds to estimating topics when abstracts are missing, and the title-only routing experiment shows that such topics can be mapped to civil subfields with moderate accuracy against independent OpenAlex labels. Even though the generic coherence gain over the concatenated control is limited, these read-outs are usable engineering outputs.

The limitations are as follows. First, all three corpora are English and the field split follows existing title/body structure; automatic field-boundary detection is untested, and the GitHub background field has a long tail (up to 20532 tokens) that the sentence-embedding baselines truncate at 128 tokens, for which the Supplementary Material reports a window-pooling control. Second, the redundancy measures are computed on two corpora with almost the same held-out predictability (ρ^2_{cca} 0.306 versus 0.245) while the ten-seed gain appears on one of them, so the two corpora define the current calibration range, and a redundancy-based pre-training criterion will need calibration on additional corpora and an independent test set. Third, the non-parametric baselines choose their own topic numbers (Top2Vec degrades to 2–4 topics on AG News) and are shown for reference only. Fourth, field-restricted read-out is available to any model with a topic–word matrix and is not unique to a two-stream encoder. Fifth, the masking rate, the whole-field dropout probability and the fusion weight are empirical values whose interaction with corpus size and field ratio is not characterised; the in-house variational baselines and FC-NTM use 20% of the training documents for early stopping (the official CombinedTM keeps its own held-out split), while the class TF–IDF read-out of FC-NTM aggregates the full training set, so the number of documents used for optimisation and for read-out differs. Sixth, all numbers come from one protocol ($K = 30$, 700-epoch budget, unified read-out and evaluation vocabulary): FC-NTM and BTM are extended to ten seeds on all three corpora (and LDA's occlusion curves for the new seeds), while the remaining baselines and the main tables stay at three seeds. Seventh, the concatenated-field control and the title-only routing task are run for three seeds per cell and against one label source (OpenAlex primary topics), so their differences should be read at the scale of the seed-to-seed variation. Finally, no independent confirmation corpus is used.

SUMMARY AND CONCLUSIONS

This paper presented FC-NTM, a field-cooperative neural topic model for dual-field short texts: two conditional variational streams share one linear topic decoder, cross-field reconstruction and a consistency regulariser encode the field relation into a shared topic space, and field-level word masking and whole-field dropout handle partial and complete field absence. A single trained model supports a decoder β read-out and a class TF–IDF read-out under one evaluation vocabulary, which removes the vocabulary and topic-length confounds of cross-model comparison.

On civil engineering title–abstract records and two public corpora the results depend on the corpus. Under the read-out-matched protocol the ten-seed comparison gives +0.028 C_V over BTM on GitHub (95% CI $[+0.012, +0.043]$, 9/10) and -0.009 on AG News (CI includes zero, 3/10); on the civil corpus both read-outs favour FC-NTM by a larger margin (+0.049 class TF–IDF and +0.061 β, 10/10), and on the two public corpora BTM keeps its lead under the native β read-out. A same-budget concatenated-field single encoder is close on every corpus (mean ΔC_V +0.010–+0.029 with mixed paired majorities), and title-only routing reaches 0.66 accuracy against independent OpenAlex labels (42.9% coverage). Two field-relation measures were tested and neither predicts the gain: the corpora have similar redundancy, and occluding the background field leaves coherence almost unchanged while changing the document-level posterior and the word-level prediction. The contribution of this paper is therefore scoped: FC-NTM provides reproducible field-restricted read-outs and missing-field inference behaviour, and read-out-matched advantages supported by ten seeds on the civil corpus and on GitHub. The complete FC-NTM configuration shows small mean coherence gains over the concatenated control, which is reported as a matched configuration-level comparison. Whether field structure is worth the extra modelling cost should be measured on the target corpus with the protocol used here rather than decided from redundancy beforehand.

DATA AVAILABILITY STATEMENT

All data, models, and code that support the findings of this study are available from the corresponding author upon reasonable request; the archived corpus splits, model checkpoints and per-seed results accompany this submission.

SUPPLEMENTAL MATERIALS

The supplementary material contains the auxiliary experiments and controls, the data and metric notes, and the case-study tables cited in the text.

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15 Title-only routing to the 15 most frequent OpenAlex topics of the civil corpus (seeds 11/23/37; 857 test records, coverage 42.9%). The TF–IDF classifier is supervised and serves as a reference point, not as a topic model. 45

TABLE 1. Topic quality on the civil engineering corpus (unified evaluation vocabulary, ten words per topic; values are means over seeds 11/23/37, and the training procedure differs across algorithm families as described in the "Baselines, training protocols and metrics" subsection; BERTopic and Top2Vec choose their own topic numbers and are shown for reference).

Model	Topics	C_V	NPMI	UMass	$\text{EXCL}_{\text{KADJ}}$	IRBO
BTM	30	0.583±0.010	0.128±0.009	−2.18	0.283	0.944
CombinedTM	30	0.539±0.029	0.082±0.028	−6.11	0.063	0.998
FC-NTM-cTFIDF	30	**0.636**±0.014	**0.159**±0.011	−2.21	0.165	0.937
FC-NTM	30	0.633±0.014	0.154±0.022	−2.85	0.014	0.990
ETM	30	0.581±0.012	0.119±0.007	−2.75	0.379	0.963
ProdLDA	30	0.580±0.023	0.125±0.017	−3.84	0.051	0.999
Top2Vec	153	0.515±0.002	0.029±0.011	−8.52	0.359	0.995
NMF	30	0.598±0.009	0.135±0.006	−2.86	0.559	0.984
BERTopic	29	0.576±0.026	0.116±0.027	−5.06	0.563	0.991
CTM-lite	30	0.536±0.030	0.098±0.027	−3.85	0.013	0.995
LDA	30	0.554±0.008	0.056±0.025	−4.62	0.602	0.987

TABLE 2. Topic quality on AG News (unified evaluation vocabulary, ten words per topic; values are means over seeds 11/23/37 where applicable, and the training procedure differs across algorithm families as described in the "Baselines, training protocols and metrics" subsection; BERTopic and Top2Vec choose their own topic numbers and are shown for reference).

Model	Topics	C_V	NPMI	UMass	$\mathrm{EXCL}_{\mathrm{KADJ}}$	IRBO
BTM	30	0.600±0.011	0.164±0.005	−4.19	0.409	0.980
CombinedTM	30	0.526±0.037	0.057±0.070	−8.01	0.118	1.000
FC-NTM-cTFIDF	30	**0.612**±0.014	**0.173**±0.012	−4.20	0.193	0.974
FC-NTM	30	0.586±0.026	0.137±0.022	−4.91	0.013	0.993
ETM	30	0.506±0.014	0.092±0.010	−5.12	0.439	0.982
ProdLDA	30	0.526±0.031	0.071±0.015	−6.61	0.055	0.999
Top2Vec	2.7	0.498±0.032	0.099±0.031	−2.92	0.268	0.937
NMF	30	0.494±0.015	0.069±0.013	−5.96	0.799	0.997
BERTopic	29	0.488±0.022	0.035±0.026	−8.81	0.681	0.997
CTM-lite	30	0.431±0.023	−0.015±0.016	−8.34	0.041	0.998
LDA	30	0.421±0.013	−0.066±0.018	−8.62	0.696	0.994

TABLE 3. Topic quality on GitHub issues (same evaluation protocol and read-outs as Table 2; the FC-NTM rows, BTM, ETM, CombinedTM, ProdLDA and CTM-lite are three-seed results).

Model	Topics	C_V	NPMI	UMass	$\text{EXCL}_{\text{KADJ}}$	IRBO
BTM	30	0.712±0.012	0.232±0.012	−1.79	0.349	0.934
FC-NTM-cTFIDF	30	**0.727**±0.022	**0.274**±0.026	−1.90	0.133	0.901
FC-NTM	30	0.701±0.016	0.233±0.012	−3.16	0.010	0.967
ETM	30	0.675±0.014	0.213±0.010	−2.14	0.366	0.955
CombinedTM	30	0.643±0.022	0.193±0.015	−4.70	0.031	0.983
NMF	30	0.614±0.005	0.164±0.003	−3.26	0.595	0.985
ProdLDA	30	0.579±0.007	0.107±0.012	−5.21	0.043	0.995
CTM-lite	30	0.585±0.032	0.129±0.036	−5.11	0.011	0.989
LDA	30	0.546±0.011	0.074±0.017	−4.21	0.684	0.990
BERTopic	29	0.540±0.015	0.076±0.012	−7.00	0.546	0.976
Top2Vec	160	0.493±0.009	−0.008±0.013	−9.65	0.337	0.994

TABLE 4. Read-out comparison ($K = 30$, means over seeds 11/23/37). Each model is first read out through its own topic–word matrix (β read-out) and then through a class TF–IDF read-out aggregated from its training-set posterior θ; BTM enters the same read-out through the point-estimate document–topic posterior of its original Section 3.3, and the FC-NTM row is the same model as in Tables 2 and 3.

Corpus	Model	β read-out C_V / NPMI	Class TF–IDF C_V / NPMI
GitHub	FC-NTM(3 seed mean)	0.701 / 0.233	**0.727 / 0.274**
	ETM	0.675 / 0.213	0.669 / 0.220
	ProdLDA	0.579 / 0.107	0.666 / 0.215
	CTM-lite	0.585 / 0.129	0.675 / 0.216
	NMF	0.614 / 0.164	0.603 / 0.158
	LDA	0.546 / 0.074	0.614 / 0.193
	BTM(3 seeds)	0.712 / 0.232	0.700 / 0.235
AG News	FC-NTM(3 seed mean)	0.586 / 0.137	**0.612 / 0.173**
	ProdLDA	0.526 / 0.071	0.544 / 0.125
	NMF	0.494 / 0.069	0.539 / 0.121
	ETM	0.506 / 0.092	0.511 / 0.082
	CTM-lite	0.431 / −0.015	0.455 / 0.062
	LDA	0.421 / −0.066	0.451 / 0.010
	BTM(3 seeds)	0.600 / 0.164	0.593 / 0.169
Civil corpus	FC-NTM(3 seed mean)	**0.633 / 0.154**	**0.636 / 0.159**
	NMF	0.598 / 0.135	0.607 / 0.142
	ETM	0.581 / 0.119	0.603 / 0.134
	BTM(3 seeds)	0.583 / 0.128	0.590 / 0.139
	LDA	0.554 / 0.056	0.575 / 0.105
	ProdLDA	0.580 / 0.125	0.545 / 0.108
	CTM-lite	0.536 / 0.098	0.530 / 0.094

TABLE 5. Ten-seed paired comparison of FC-NTM and BTM on the three corpora. The read-out rows report C_V differences (FC-NTM minus BTM at the same read-out); the occlusion rows compare the relative degradation NLL(100%)/NLL(25%), so a positive value means FC-NTM is more stable. The occlusion rows use the pre-selected baselines (BTM on all three corpora and additionally LDA on AG News) rather than a full baseline tournament. CIs are two-sided t intervals; sign and Wilcoxon tests are two-sided.

Corpus	Comparison	Mean diff.	95% CI low	95% CI high	Positive	Sign p	Wilcoxon p
GitHub	Class TF–IDF	+0.028	+0.012	+0.043	9/10	0.021	0.004
GitHub	β read-out	−0.014	−0.028	−0.001	3/10	0.344	0.037
GitHub	Occlusion (vs. BTM)	+0.024	+0.022	+0.026	10/10	0.002	0.002
AG News	Class TF–IDF	−0.009	−0.030	+0.013	3/10	0.344	0.375
AG News	β read-out	−0.027	−0.047	−0.008	1/10	0.021	0.004
AG News	Occlusion (vs. BTM)	+0.025	+0.024	+0.026	10/10	0.002	0.002
AG News	Occlusion (vs. LDA)	−0.013	−0.014	−0.012	0/10	0.002	0.002
Civil corpus	Class TF–IDF	+0.049	+0.033	+0.064	10/10	0.002	0.002
Civil corpus	β read-out	+0.061	+0.047	+0.075	10/10	0.002	0.002
Civil corpus	Occlusion (vs. BTM)	−0.003	−0.004	−0.002	0/10	0.002	0.002

TABLE 6. Field redundancy, measured occlusion gap and the C_V difference against BTM on two corpora. Redundancy is fitted on the training split and evaluated on the test split and includes constructive controls (exact copy = 1.000, cross-document shuffle ≈ 0.000); the occlusion ΔC_V is the mean±standard deviation over seeds 11/23/37 with dropout training. The last column gives the C_V difference under the matched class TF–IDF read-out and under BTM's native β read-out.

Corpus	Held-out Ridge R^2	Held-out ρ^2_{cca}	Occlusion ΔC_V	C_V gap, cTFIDF−BTM (matched / β)
GitHub issues	0.230	0.306	+0.009±0.022	+0.027 / +0.014
AG News	0.158	0.245	+0.018±0.013	+0.019 / +0.012

TABLE 7. Partial-occlusion curves on GitHub issues (seed 11). NLL is the per-token negative log-likelihood of the occluded background tokens (lower is better); relative degradation is NLL(100%)/NLL(25%); agreement is the fraction of documents whose most probable topic is unchanged at 100% occlusion relative to the unoccluded ($p = 0$) inference. BTM enters the protocol through the document–topic posterior of its Section 3.3.

Model	NLL(25%)	NLL(50%)	NLL(75%)	NLL(100%)	Relative degradation / Agreement(100%)
FC-NTM	7.712	7.735	7.790	7.960	**1.032** / 0.358
LDA	6.802	6.854	6.948	7.302	1.074 / 0.420
ProdLDA	7.558	7.588	7.668	8.165	1.080 / 0.354
BTM	6.630	6.639	6.670	**6.972**	1.052 / 0.448
ETM	**6.616**	6.643	6.708	7.202	1.089 / 0.452
NMF	6.972	7.169	7.561	10.255	1.471 / **0.469**

TABLE 8. Partial-occlusion behaviour on the three corpora (seeds 11/23/37, mean±standard deviation; the mask is redrawn per seed). Bold marks the smallest relative degradation per corpus. Field asymmetry is described by the mean background/summary token ratio and by the token-type share of background words that do not occur in the summary of the same document.

Corpus(Field asymmetry)	Model	Relative degradation	Agreement(100%)
GitHub issues Length ratio 12.9 Background-exclusive 0.877	FC-NTM	**1.029±0.003**	0.345
	LDA	1.074±0.000	0.424
	ProdLDA	1.081±0.002	0.347
	ETM	1.088±0.003	0.455
	BTM	1.052±0.001	0.438
	NMF	1.470±0.006	0.467
Civil corpus Length ratio 11.2 Background-exclusive 0.894	FC-NTM	1.010±0.001	0.673
	LDA	1.036±0.002	0.561
	ProdLDA	1.027±0.003	0.552
	ETM	1.041±0.001	0.655
	BTM	**1.007±0.001**	0.561
	NMF	1.173±0.001	0.655
AG News Length ratio 3.9 Background-exclusive 0.868	FC-NTM	1.018±0.001	0.520
	LDA	**1.005±0.001**	0.436
	ProdLDA	1.026±0.001	0.489
	ETM	1.072±0.001	0.554
	BTM	1.044±0.001	0.555
	NMF	1.408±0.003	0.546

TABLE 9. Field-restricted read-out (C_V, mean±standard deviation over seeds 11/23/37). Every topic–word matrix is truncated to the vocabulary of one field and scored against that field's test text only; "background view" is the abstract/body side and "summary view" the title side. FC-NTM's summary view is scored on 20–22 scorable topics per seed on GitHub (29–30 on AG News, 26–28 on the civil corpus) because topics whose ten field words never occur in the held-out text get an undefined C_V.

Corpus	Model	Background view C_V	Summary view C_V
GitHub issues	FC-NTM	0.697±0.016	0.426±0.012
	BTM	**0.710±0.011**	0.396±0.018
	NMF	0.609±0.004	0.433±0.002
	LDA	0.543±0.011	0.466±0.005
	ProdLDA	0.575±0.006	**0.467±0.034**
AG News	FC-NTM	0.566±0.016	0.423±0.015
	BTM	**0.586±0.015**	0.399±0.018
	NMF	0.477±0.015	0.444±0.002
	LDA	0.419±0.007	**0.486±0.006**
	ProdLDA	0.499±0.032	0.438±0.007
Civil corpus	FC-NTM	**0.637±0.015**	0.408±0.005
	BTM	0.589±0.014	0.382±0.008
	NMF	0.600±0.009	0.398±0.013
	LDA	0.554±0.008	0.415±0.006
	ProdLDA	0.576±0.025	**0.419±0.012**

TABLE 10. Three-step fusion ablation on GitHub issues ($K = 30$, seeds 11/23/37, same protocol). Within one read-out each row changes exactly one thing: first the fused mixture θ_{fuse} is decoded into the reconstruction term (mean fusion), then mean fusion is replaced by a learnable gate (the main model). "Positive pairs" counts the seeds whose paired difference against the row above is positive (C_V, NPMI).

Read-out	Configuration	C_V	NPMI	Positive pairs	UMass
β	No fusion	0.721±0.016	0.266±0.012	—	−2.63
	Mean fusion + fusion recon.	0.708±0.016	0.245±0.013	1/3, 0/3	−2.98
	Gated fusion (main model)	0.701±0.016	0.233±0.012	0/3, 0/3	−3.16
cTFIDF	No fusion	0.714±0.001	0.262±0.007	—	−1.86
	Mean fusion + fusion recon.	0.728±0.007	0.275±0.015	3/3, 3/3	−1.91
	Gated fusion (main model)	0.727±0.022	0.274±0.026	1/3, 1/3	−1.90

TABLE 11. Core-component ablation on three corpora and two read-outs ($K = 30$, seeds 11/23/37, 700-epoch budget, all other settings identical to the main model). Each column switches off one component: cross-field reconstruction of both streams, the JS consistency weight, field-level word masking, and whole-field dropout; the fusion-reconstruction term is kept in all columns. Values are mean paired ΔC_V (negative means the component matters), parentheses count positive pairs.

Corpus	Read-out	−Cross-field recon.	−JS consistency	−Word masking	−Field dropout
GitHub	β	+0.021 (2/3)	−0.001 (1/3)	−0.012 (1/3)	−0.023 (1/3)
GitHub	Class TF-IDF	−0.005 (1/3)	+0.004 (2/3)	−0.004 (1/3)	−0.021 (1/3)
AG News	β	−0.011 (2/3)	+0.003 (1/3)	−0.026 (0/3)	−0.010 (1/3)
AG News	Class TF-IDF	−0.022 (0/3)	−0.002 (1/3)	−0.043 (0/3)	−0.023 (1/3)
Civil corpus	β	−0.022 (0/3)	+0.001 (2/3)	−0.010 (1/3)	+0.001 (1/3)
Civil corpus	Class TF-IDF	−0.019 (0/3)	−0.002 (0/3)	−0.012 (0/3)	−0.034 (0/3)

TABLE 12. Sensitivity to the fusion-reconstruction weight λ_f on GitHub issues (gated fusion, seeds 11/23/37). $\lambda_f = 0$ is the no-fusion setting, $\lambda_f = 0.5$ is the value used throughout; all other settings match the main model.

λ_f	β read-out		Class TF–IDF	
	C_V	NPMI	C_V	NPMI
0	0.721±0.016	0.266±0.012	0.714±0.001	0.262±0.007
0.25	0.697±0.015	0.217±0.031	0.713±0.015	0.262±0.004
0.5	0.701±0.016	0.233±0.012	0.727±0.022	0.274±0.026
1.0	0.716±0.012	0.246±0.026	0.723±0.012	0.277±0.018

TABLE 13. Whole-field occlusion at inference time ($K = 30$, final configuration, 700-epoch budget). ΔC_V is the coherence difference between full-field and occluded inference under the same read-out; ΔNLL is the per-token increase; agreement is the fraction of documents whose most probable topic is unchanged. The two rows of a corpus use the same seeds, isolating the effect of the training-time dropout setting.

Corpus	Training setting	C_V^{full}	C_V^{occ}	ΔC_V	ΔNLL(per token)	Agreement	Mean JS
GitHub	Dropout ($p = 0.5$)	0.727	0.717	+0.009	+0.241	0.345	0.082
	No dropout	0.705	0.725	−0.020	+0.242	0.346	0.092
Civil corpus	Dropout ($p = 0.5$)	0.636	0.627	+0.009	+0.103	0.673	0.033
	No dropout	0.602	0.597	+0.005	+0.125	0.642	0.039
AG News	Dropout ($p = 0.5$)	0.612	0.594	+0.018	+0.232	0.520	0.049
	No dropout	0.589	0.582	+0.007	+0.269	0.498	0.060

TABLE 14. Concatenated-field single-encoder control (seeds 11/23/37; same budget, vocabulary and read-outs). Δ is FC-NTM minus the control, with positive pairs in parentheses.

Corpus	Read-out	FC-NTM	Concatenated	Δ (positive pairs)
GitHub	β	0.701±0.016	0.683±0.014	+0.017 (2/3)
GitHub	Class TF–IDF	0.727±0.022	0.710±0.020	+0.017 (1/3)
AG News	β	0.586±0.026	0.574±0.019	+0.012 (2/3)
AG News	Class TF–IDF	0.612±0.014	0.584±0.009	+0.028 (3/3)
Civil corpus	β	0.633±0.014	0.623±0.014	+0.010 (2/3)
Civil corpus	Class TF–IDF	0.636±0.014	0.606±0.014	+0.029 (2/3)

TABLE 15. Title-only routing to the 15 most frequent OpenAlex topics of the civil corpus (seeds 11/23/37; 857 test records, coverage 42.9%). The TF–IDF classifier is supervised and serves as a reference point, not as a topic model.

Method	Input	Accuracy	Macro-F1
TF–IDF logistic regression (supervised)	title	0.810	0.805
FC-NTM	title + abstract	0.719±0.016	0.676±0.032
FC-NTM	title only	0.664±0.004	0.624±0.013
Concatenated control	title only	0.653±0.012	0.614±0.034
BTM	title only	0.600±0.013	0.529±0.043
LDA	title only	0.532±0.039	0.458±0.057

List of Figures

1 Main results on the two public corpora (unified evaluation vocabulary, ten words per topic). Left: C_V; right: NPMI. Error bars are standard deviations across seeds. The FC-NTM rows, BTM and ETM on GitHub and the FC-NTM rows on AG News are three-seed results; CombinedTM, ProdLDA and CTM-lite were run for the same seeds 11/23/37 under the same protocol. . . 48

2 Field redundancy and gain (held-out measures). Left: squared canonical correlation ρ^2_{cca} (higher means more redundant; fitted on the training split and evaluated on the test split); middle: the ΔC_V of the real occlusion diagnostic (GitHub $+0.009 \pm 0.022$, AG News $+0.018 \pm 0.013$, three seeds each); right: the C_V difference between FC-NTM-cTFIDF and BTM under the matched read-out (GitHub $+0.027$, AG News $+0.019$). The two corpora have similar redundancy, while the ten-seed gain is positive on GitHub only (AG News -0.009), so the figure does not support a redundancy-based prediction of the gain. 49

3 Inference behaviour under partial occlusion of the background field (GitHub issues, seed 11, 2000 test documents). Left: per-token negative log-likelihood of the occluded tokens; right: relative degradation NLL(p)/NLL(0.25), which summarises the relative increase from the 25% condition without removing calibration differences. FC-NTM is trained with whole-field dropout at $p = 0.5$ and has the smallest relative degradation ($\times 1.032$); the ratio itself cannot separate "recovering content from the summary" from "being insensitive to the input", and the absolute NLL and agreement are not the best, as reported in the text. 50

4 Relative degradation under the same partial-occlusion protocol on three corpora (curves are means, shading is ±1 standard deviation; FC-NTM, BTM and LDA cover the ten seeds 11–139, the remaining baselines keep the archived seeds 11/23/37). Left: GitHub issues (background to summary length ratio ≈ 12.9); middle: civil engineering corpus (≈ 11.2); right: AG News (≈ 3.9). Only on GitHub is the degradation of FC-NTM clearly below all baselines; on the civil corpus and AG News the smallest degradation belongs to BTM and LDA respectively, and the gap to FC-NTM is within 0.003–0.013. 51

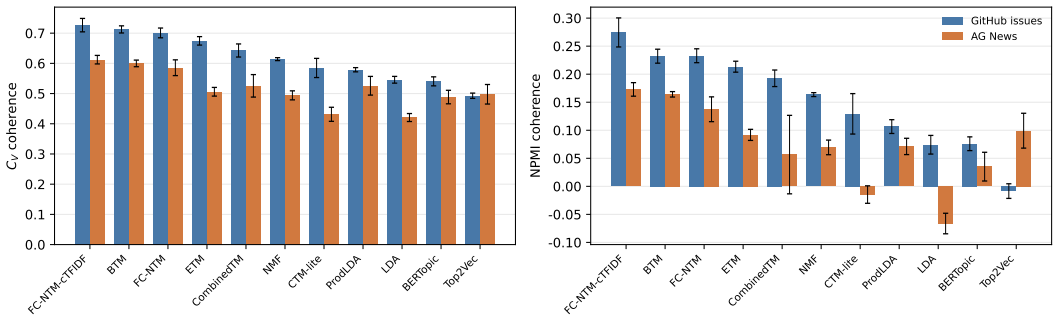


Fig. 1. Main results on the two public corpora (unified evaluation vocabulary, ten words per topic). Left: C_V; right: NPMI. Error bars are standard deviations across seeds. The FC-NTM rows, BTM and ETM on GitHub and the FC-NTM rows on AG News are three-seed results; CombinedTM, ProdLDA and CTM-lite were run for the same seeds 11/23/37 under the same protocol.

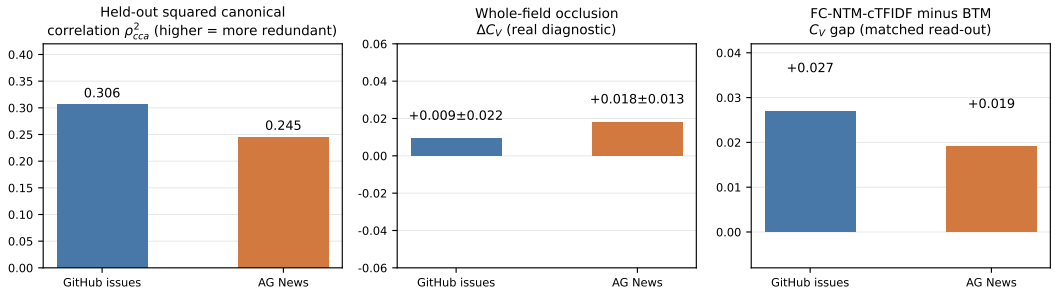


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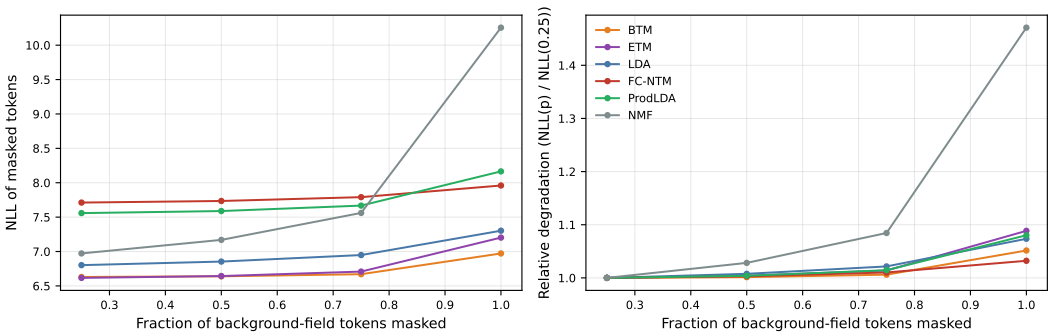


Fig. 3. Inference behaviour under partial occlusion of the background field (GitHub issues, seed 11, 2000 test documents). Left: per-token negative log-likelihood of the occluded tokens; right: relative degradation NLL(p)/NLL(0.25), which summarises the relative increase from the 25% condition without removing calibration differences. FC-NTM is trained with whole-field dropout at $p = 0.5$ and has the smallest relative degradation ($\times 1.032$); the ratio itself cannot separate "recovering content from the summary" from "being insensitive to the input", and the absolute NLL and agreement are not the best, as reported in the text.

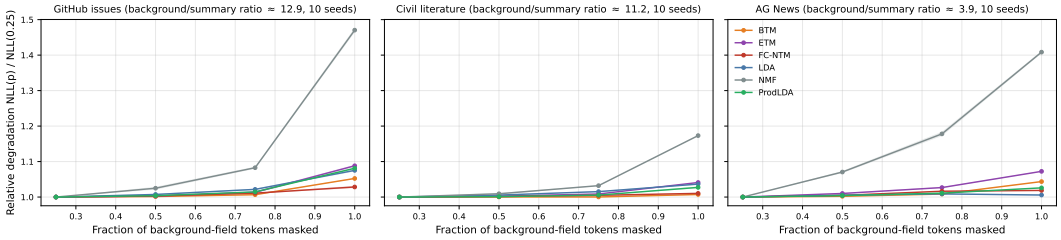


Fig. 4. Relative degradation under the same partial-occlusion protocol on three corpora (curves are means, shading is ±1 standard deviation; FC-NTM, BTM and LDA cover the ten seeds 11–139, the remaining baselines keep the archived seeds 11/23/37). Left: GitHub issues (background to summary length ratio ≈ 12.9); middle: civil engineering corpus (≈ 11.2); right: AG News (≈ 3.9). Only on GitHub is the degradation of FC-NTM clearly below all baselines; on the civil corpus and AG News the smallest degradation belongs to BTM and LDA respectively, and the gap to FC-NTM is within 0.003–0.013.